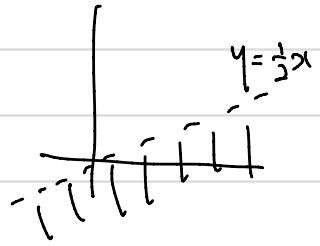


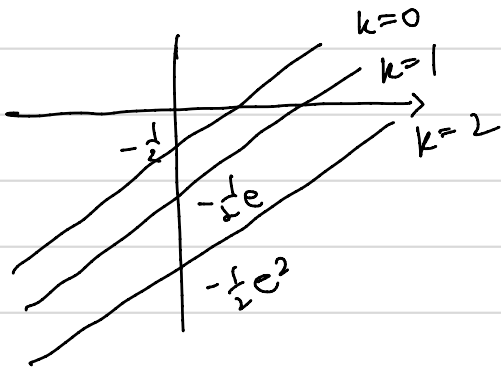
1. $f(x, y) = \ln(x - 2y)$

(a) Domain : $\{ (x, y) : x - 2y > 0 \}$
 range : $\mathbb{R}$



(b) $k = \ln(x - 2y) \Leftrightarrow x - 2y = e^k$

$\Leftrightarrow y = \frac{1}{2}x - \frac{1}{2}e^k$



(c) u is the unit vector in the direction of the $\nabla f(x, y) = \left(\frac{1}{x-2y}, -\frac{2}{x-2y} \right)$

$\therefore u = \frac{\nabla f(x, y)}{|\nabla f(x, y)|} = \left(\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right)$

(d) $f_x(3, 1) = 1, f_y = -2$

① tangent plane : $z = 1(x-3) - 2(y-1) + 0$

$\Leftrightarrow x - 2y - z - 1 = 0$

② normal line : Since a directional vector

$\langle 1, -2, -1 \rangle$,

$\frac{x-3}{1} = \frac{y-1}{-2} = \frac{z}{-1}$

$$2. (a) f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = 0$$

$$f_y(0,0) = \lim_{h \rightarrow 0} \frac{f(0,h) - f(0,0)}{h} = 0.$$

$$(b) D_{\mathbf{u}} f(0,0) = \lim_{h \rightarrow 0} \frac{f(ha, hb) - f(0,0)}{h}$$

$$U = \langle a, b \rangle$$

$$= \lim_{h \rightarrow 0} \frac{h^3 a b^2}{h (h^2 a^2 + h^4 b^4)}$$

$$= \lim_{h \rightarrow 0} \frac{a b^2}{(a^2 + h^2 b^4)} = \frac{b^2}{a} //$$

$$(c) \text{ If } (x,y) \neq (0,0),$$

$$f_x(x,y) = \frac{y^2(x^2+y^4) - xy^2(2x)}{(x^2+y^4)^2}$$

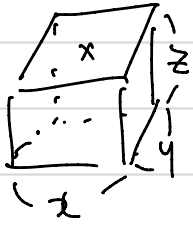
$$= \frac{-x^2y^2 + y^6}{(x^2+y^4)^2}$$

$$\text{Then } \lim_{(x,y) \xrightarrow{x \rightarrow 0} (0,0)} f_x(x,y)$$

$$= \lim_{\substack{y \rightarrow 0 \\ x=0}} \frac{y^6}{y^8} = \lim_{y \rightarrow 0} \frac{1}{y^2} = \infty$$

$\therefore f_x$ is not continuous at $(0,0)$

3.



$$xyz = 4 \quad (x, y, z > 0)$$

$$\text{Area} : xy + 2yz + 2zx = xy + 8\left(\frac{1}{x} + \frac{1}{y}\right) = f(x, y)$$

① Critical points of f

$$f_x = y - \frac{8}{x^2} = 0 \Rightarrow x^2 y = 8$$

$$f_y = x - \frac{8}{y^2} = 0 \Rightarrow x = \frac{8}{y^2}$$

$$\text{so } \frac{64}{y^3} = 8 \Rightarrow y^3 - 8 = 0$$

$$\Rightarrow y = 2 \text{ \& } x = 2$$

so $(2, 2)$ is a critical point

② Second derivative test.

$$f_{xx} = \frac{16}{x^3}, \quad f_{yy} = \frac{16}{y^3}, \quad f_{xy} = 1$$

$$(i) f_{xx}(2, 2) = 2 > 0$$

$$(ii) f_{xx}(2, 2) f_{yy}(2, 2) - f_{xy}(2, 2)^2$$

$$= 2 \cdot 2 - 1 > 0.$$

so f has a local minimum at $(2, 2)$

But f has only one critical point

This is an absolute minimum and

$$f(2, 2) = 12$$

4. Use Lagrange multipliers. Let $g(x, y, z) = x - y$
 $f(x, y, z) = x^2 + y^2 + z^2$ and $h(x, y, z) = y^2 + z^2$

$$\begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z) + \mu \nabla h(x, y, z) \\ g(x, y) = \frac{1}{2} \\ h(x, y) = 1 \end{cases}$$

$$\Rightarrow \begin{cases} 2x = \lambda & (1) \\ 2y = -\lambda + 2y\mu & (2) \\ 2z = 2\mu z & (3) \\ x - y = \frac{1}{2} & (4) \\ z^2 + y^2 = 1 & (5) \end{cases}$$

(i) $z \neq 0$ By (3), $\mu = 1 \Rightarrow$ By (2), $\lambda = 0$
 $\Rightarrow$ By (1), $x = 0 \Rightarrow$ By (4), $y = -\frac{1}{2}$
 $\Rightarrow$ By (5), $z = \pm \frac{\sqrt{3}}{2}$

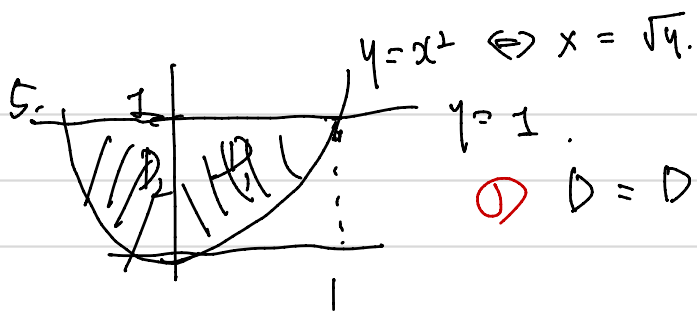
$$\therefore f\left(0, -\frac{1}{2}, \pm \frac{\sqrt{3}}{2}\right) = 1$$

(ii) $z = 0$ By (5), $y = \pm 1$
 By (4), $x = \frac{1}{2} \pm 1$

$$f\left(\frac{3}{2}, 1, 0\right) = \frac{13}{4}, \quad f\left(-\frac{1}{2}, -1, 0\right) = \frac{5}{4}$$

$(\lambda = 3, \mu = \frac{5}{2})$
 $(\lambda = -1, \mu = \frac{1}{2})$

Maximum : $\frac{13}{4}$. Minimum : 1



The domain D could be

$$\textcircled{1} D = D_1 = \{ (x, y) \mid 0 \leq x \leq 1, x^2 \leq y \leq 1 \}$$

$$= \{ (x, y) \mid 0 \leq y \leq 1, 0 \leq x \leq \sqrt{y} \}$$

$$\textcircled{2} D = D_1 \cup D_2 = \{ (x, y) \mid 0 \leq y \leq 1, -\sqrt{y} \leq x \leq \sqrt{y} \}$$

$$\textcircled{3} D = D_2 = \{ (x, y) \mid 0 \leq y \leq 1, \sqrt{y} \leq x \leq 0 \}$$

$$\textcircled{1}: \iint_D e^{y^{3/2}} dA = \int_0^1 \int_0^{\sqrt{y}} e^{y^{3/2}} dx dy = \int_0^1 \sqrt{y} e^{y^{3/2}} dy$$

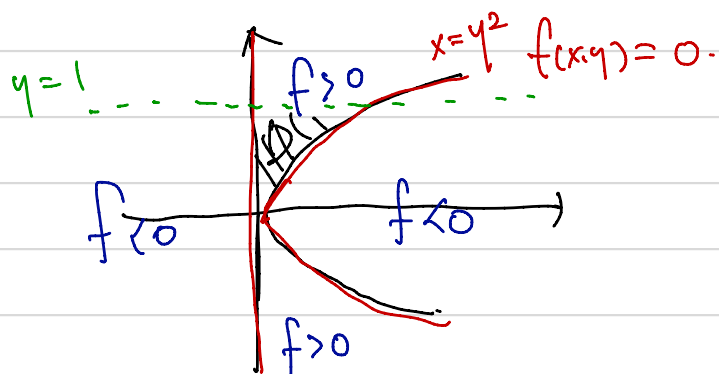
$$= \frac{2}{3} [e^{y^{3/2}}]_0^1 = \frac{2}{3} (e - 1)$$

$$\textcircled{2}: \iint_D e^{y^{3/2}} dA = \int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} e^{y^{3/2}} dx dy = 2 \int_0^1 \int_0^{\sqrt{y}} e^{y^{3/2}} dx dy$$

$$\textcircled{3}: \iint_D e^{y^{3/2}} dA = \int_0^1 \int_{-\sqrt{y}}^0 e^{y^{3/2}} dx dy = \frac{2}{3} (e - 1) = \frac{4}{3} (e - 1)$$

6. Let $f(x, y) = -x^2 + xy^2$.

Note $0 = f(x, y) = -x^2 + xy^2 = x(-x + y^2)$.



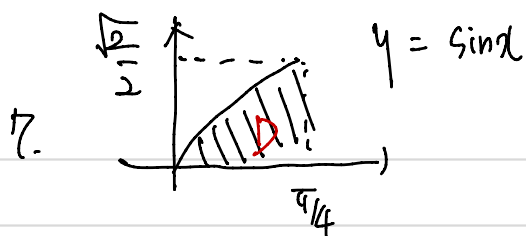
$$\therefore V = \iint_D f(x, y) dA$$

$$= \int_0^1 \int_0^{y^2} (-x^2 + xy^2) dx dy$$

$$(D = \{ (x, y) : 0 \leq y \leq 1, 0 \leq x \leq y^2 \})$$

$$= \int_0^1 \left(-\frac{1}{3} y^6 + \frac{1}{2} y^6 \right) dy$$

$$= \int_0^1 \frac{1}{6} y^6 dy = \frac{1}{42}$$



$$D = \{(x, y) \mid 0 \leq x \leq \frac{\pi}{4}, 0 \leq y \leq \sin x\}$$

$$= \{(x, y) \mid 0 \leq y \leq \frac{\sqrt{2}}{2}, \arcsin y \leq x \leq \frac{\pi}{4}\}$$

Sol 1

$$\int_0^{\frac{\sqrt{2}}{2}} \int_{\arcsin y}^{\frac{\pi}{4}} \cos x \sqrt{1 - \cos^2 x} \, dx \, dy$$

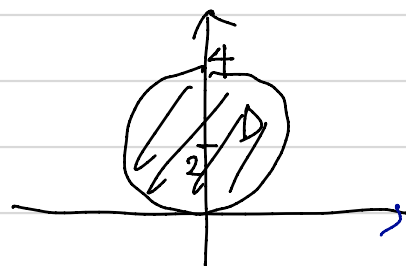
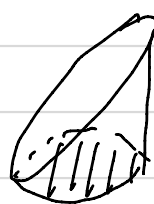
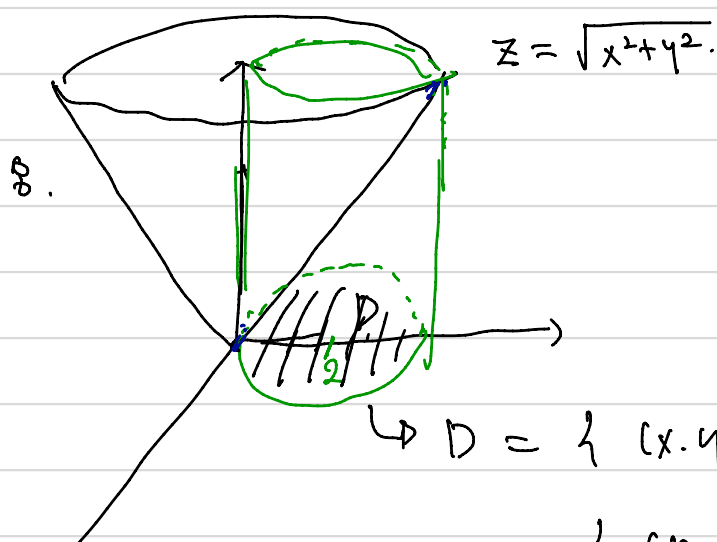
$$= \int_0^{\frac{\sqrt{2}}{2}} \int_{\arcsin y}^{\frac{\pi}{4}} \cos x \sin x \, dx \, dy = \int_0^{\frac{\sqrt{2}}{2}} \left[\frac{1}{2} \sin^2 x \right]_{x=\arcsin y}^{x=\frac{\pi}{4}} \, dy$$

$$= \int_0^{\frac{\sqrt{2}}{2}} \frac{1}{4} - \frac{1}{2} y^2 \, dy = \frac{\sqrt{2}}{8} - \frac{1}{6} \frac{\sqrt{2}}{4} = \frac{\sqrt{2}}{12}$$

Sol 2

$$\int_0^{\frac{\pi}{4}} \int_0^{\sin x} \cos x \sqrt{1 - \cos^2 x} \, dy \, dx = \int_0^{\frac{\pi}{4}} \sin x \cos x \sqrt{1 - \cos^2 x} \, dx$$

$$= \frac{1}{2} \left[\frac{2}{3} (1 - \cos^2 x)^{\frac{3}{2}} \right]_0^{\frac{\pi}{4}} = \frac{1}{3} \left(1 - \frac{1}{2} \right)^{\frac{3}{2}} = \frac{\sqrt{2}}{12}$$



$$D = \{(x, y) : x^2 + y^2 \leq 4\}$$

$$= \{(r, \theta) : 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi\}$$

Sol

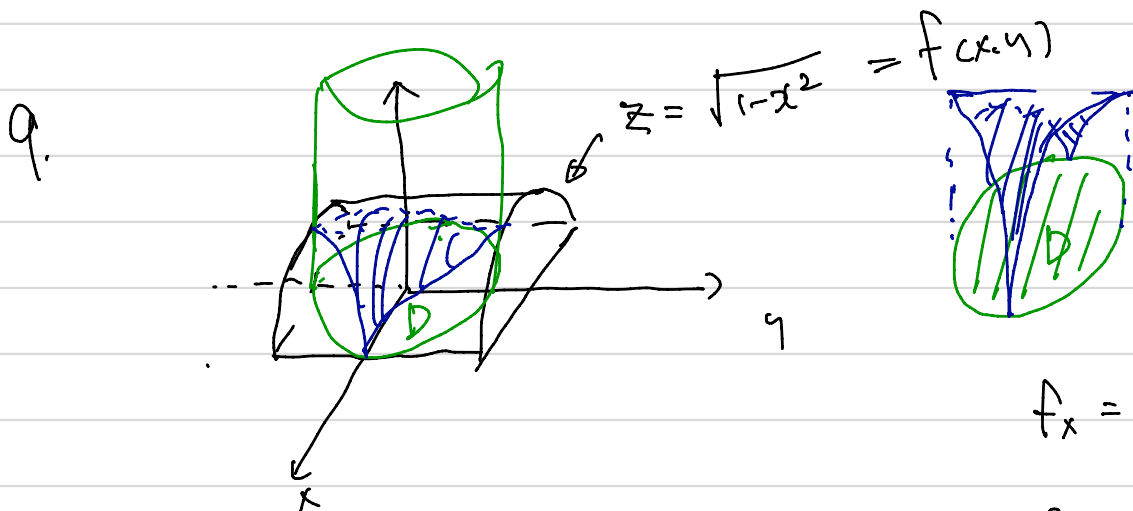
$$V = \iint_D \sqrt{x^2 + y^2} \, dA = \int_0^{2\pi} \int_0^2 r \, r \, dr \, d\theta$$

$$= \int_0^\pi \int_0^{4\sin\theta} r^2 dr d\theta = \int_0^\pi \frac{1}{3} (4\sin\theta)^3 d\theta$$

$$= \frac{64}{3} \int_0^\pi \sin\theta - \cos^2\theta \sin\theta d\theta$$

$$= \frac{64}{3} \left(2 + \left[\frac{1}{3} \cos^3\theta \right]_0^\pi \right)$$

$$= \frac{64}{3} \left(2 - \frac{2}{3} \right) = \frac{256}{9} "$$



$$f_x = \frac{-x}{\sqrt{1-x^2}}$$

$$f_y = 0$$

$$A = \iint_D \sqrt{f_x^2 + f_y^2 + 1} dA$$

$$D = \{(x,y) \mid x^2 + y^2 \leq 1\}$$

$$= \iint_D \sqrt{\frac{1}{1-x^2}} dA =$$

Sol 1 //

$$\iint_D \sqrt{\frac{1}{1-x^2}} dA = \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\sqrt{1-x^2}} dy dx$$

$$= 2 \int_{-1}^1 1 dy = 4 "$$

$$\underline{\text{Sol 2}} \quad \iint_D \sqrt{\frac{1}{1-x^2}} dA = 4 \int_0^{\frac{\pi}{2}} \int_0^1 \sqrt{\frac{1}{1-r^2 \cos^2 \theta}} r dr d\theta$$

$$= 4 \int_0^{\frac{\pi}{2}} \frac{1}{-2 \cos^2 \theta} \left[2 (1-r^2 \cos^2 \theta)^{\frac{1}{2}} \right]_{r=0}^{r=1} d\theta$$

$$= 4 \int_0^{\frac{\pi}{2}} \frac{1}{\cos^2 \theta} (1 - (1 - \cos^2 \theta)^{\frac{1}{2}}) d\theta$$

$$= 4 \int_0^{\frac{\pi}{2}} \frac{1 - \sin \theta}{\cos^2 \theta} d\theta = 4 \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta} d\theta$$

$$= 4 \left[-\frac{2}{1 + \tan(\theta/2)} \right]_0^{\frac{\pi}{2}}$$

$$= 8 \left(-\frac{1}{2} + 1 \right) = 4.$$